MRC Lecture 1 — Derivation Guide 3

Cosine Similarity and L2 Normalization

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1. Motivation

In Derivation Guides 1 and 2, we established that the dot product measures compatibility and that matrix multiplication computes all pairwise compatibilities simultaneously. There is, however, a problem with using the raw dot product as a similarity measure.

The dot product conflates two distinct quantities: the **alignment** between two vectors (the angle between them) and their **magnitudes** (how long they are). A vector with very large magnitude will produce a large dot product with almost any other vector, regardless of direction. In molecular terms, this would mean that a molecule with extremely strong features in every dimension would appear "compatible" with everything — not because of genuine complementarity, but because of magnitude alone.

We want a similarity measure that captures pure alignment — just the angle — independent of magnitude. **Cosine similarity** achieves this, and **L2 normalization** is the operation that makes cosine similarity equivalent to the dot product.

2. Definition of Cosine Similarity

Definition. The cosine similarity between two non-zero vectors $a, b \in \mathbb{R}^n$ is:

$$\text{sim}(a,b) = \frac{a \cdot b}{\| a \| \| b \|}$$

From the geometric definition of the dot product (Derivation Guide 1, §4), we know that $a \cdot b = \| a \| \| b \| \cos\theta$. Substituting:

$$\text{sim}(a,b) = \frac{\| a \| \| b \| \cos\theta}{\| a \| \| b \|} = \cos\theta$$

Cosine similarity is the cosine of the angle between two vectors. The magnitudes cancel completely.

Range. Since $\cos\theta \in [-1, +1]$:

- $\text{sim}(a,b) = +1$: vectors point in the same direction ($\theta = 0°$)
- $\text{sim}(a,b) = 0$: vectors are orthogonal ($\theta = 90°$)
- $\text{sim}(a,b) = -1$: vectors point in opposite directions ($\theta = 180°$)

3. Why Cosine Similarity Removes Magnitude Dependence

Claim. Cosine similarity depends only on the direction of the vectors, not on their magnitudes. Formally: for any positive scalars $\alpha, \beta > 0$,

$$\text{sim}(\alpha a, \beta b) = \text{sim}(a,b)$$

Proof.

$$\text{sim}(\alpha a, \beta b) = \frac{(\alpha a) \cdot (\beta b)}{\| \alpha a \| \| \beta b \|}$$

Step 1. Expand the numerator using bilinearity of the dot product:

$$(\alpha a) \cdot (\beta b) = \alpha\beta (a \cdot b)$$

Step 2. Expand the denominator using the property $\| \alpha a \| = |\alpha| \| a \| = \alpha \| a \|$ (since $\alpha > 0$):

$$\| \alpha a \| \| \beta b \| = \alpha \| a \| \cdot \beta \| b \| = \alpha\beta \| a \| \| b \|$$

Step 3. Divide:

$$\mathrm{sim}(\alpha a, \beta b) = \frac{\alpha\beta(a \cdot b)}{\alpha\beta \| a \| \| b \|} = \frac{a \cdot b}{\| a \| \| b \|} = \mathrm{sim}(a, b) \quad \blacksquare$$

Scaling either vector by any positive constant does not change the cosine similarity. Only the directions matter.

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4. Worked Example 1: Cosine Similarity Computation

Problem. Compute the cosine similarity between $a = (2, -1, 3)$ and $b = (1, -2, 2)$.
Step 1. Compute the dot product:

$$a \cdot b = 2(1) + (-1)(-2) + 3(2) = 2 + 2 + 6 = 10$$

Step 2. Compute the norms:

$$\| a \| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{4 + 1 + 9} = \sqrt{14} \approx 3.742$$
$$\| b \| = \sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

Step 3. Compute cosine similarity:

$$\mathrm{sim}(a, b) = \frac{10}{\sqrt{14} \times 3} = \frac{10}{3\sqrt{14}} = \frac{10}{11.225} \approx 0.891$$

Interpretation. The cosine similarity is 0.891 — close to 1, indicating strong alignment. These vectors point in nearly the same direction. In a molecular embedding space, this would indicate that the two molecules are highly compatible — they are likely to bind.

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5. Worked Example 2: Magnitude Independence

Problem. Verify that scaling does not change cosine similarity. Compute $\mathrm{sim}(10a, 0.5b)$ using the same vectors from Example 1.
Step 1. Scale the vectors:

$$10a = (20, -10, 30), 0.5b = (0.5, -1, 1)$$

Step 2. Compute the dot product:

$$10a \cdot 0.5b = 20(0.5) + (-10)(-1) + 30(1) = 10 + 10 + 30 = 50$$

Step 3. Compute the norms:

$$\| 10a \| = 10 \| a \| = 10\sqrt{14} \approx 37.42$$
$$\| 0.5b \| = 0.5 \| b \| = 0.5 \times 3 = 1.5$$

Step 4. Compute cosine similarity:

$$\mathrm{sim}(10a, 0.5b) = \frac{50}{37.42 \times 1.5} = \frac{50}{56.13} \approx 0.891$$

The result is identical to Example 1. The magnitudes changed by factors of 10 and 0.5 respectively, but the cosine similarity is unchanged.

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6. L2 Normalization

Definition. The L2 normalization of a non-zero vector a is:

$$\hat{a} = \frac{a}{\| a \|}$$

The resulting vector $\hat{a}$ is called the **unit vector** in the direction of a.
Property. The unit vector has length 1:

$$\| \hat{a} \| = \left\| \frac{a}{\| a \|} \right\| = \frac{\| a \|}{\| a \|} = 1$$

Geometric interpretation. L2 normalization projects every vector onto the **unit hypersphere** — the surface of the sphere with radius 1 centered at the origin. In 2D, this is the unit circle. In 3D, this is the unit sphere. In d dimensions, this is the d-dimensional hypersphere S^{d-1}. After normalization, all vectors have the same magnitude (1) and differ only in their direction.

7. The Key Identity: After L2 Normalization, Dot Product Equals Cosine Similarity

Claim. If $\hat{a}$ and $\hat{b}$ are L2-normalized vectors, then:

$$\hat{a} \cdot \hat{b} = \text{sim}(a, b) = \cos\theta$$

Proof.

$$\hat{a} \cdot \hat{b} = \frac{a}{\|a\|} \cdot \frac{b}{\|b\|} = \frac{a \cdot b}{\|a\|\|b\|} = \text{sim}(a, b) = \cos\theta \ \blacksquare$$

Why this matters. After L2 normalization, the simple dot product — the cheapest similarity operation — gives exactly the cosine similarity — the magnitude-independent similarity measure. There is no computational cost to removing magnitude dependence; we simply normalize the vectors once and then use dot products everywhere.

This is precisely what CLIP (Radford et al., 2021) and CALM (Lee et al., 2026) do: L2-normalize all embeddings, then compute pairwise similarities via $S = QK^T$ (Derivation Guide 2). Every entry $S_{ij} = \hat{q}_i \cdot \hat{k}_j = \cos\theta_{ij}$.

8. Worked Example 3: L2 Normalization and Dot Product

Problem. L2-normalize $a = (2, -1, 3)$ and $b = (1, -2, 2)$, then compute their dot product.

Step 1. Normalize a:

$$\|a\| = \sqrt{14}, \hat{a} = \frac{1}{\sqrt{14}}(2, -1, 3) = (0.535, -0.267, 0.802)$$

Step 2. Normalize b:

$$\|b\| = 3, \hat{b} = \frac{1}{3}(1, -2, 2) = (0.333, -0.667, 0.667)$$

Step 3. Verify unit length:

$$\|\hat{a}\| = \sqrt{0.535^2 + 0.267^2 + 0.802^2} = \sqrt{0.286 + 0.071 + 0.643} = \sqrt{1.000} = 1 \checkmark$$
$$\|\hat{b}\| = \sqrt{0.333^2 + 0.667^2 + 0.667^2} = \sqrt{0.111 + 0.445 + 0.445} = \sqrt{1.001} \approx 1 \checkmark$$

(The small deviation from 1 is due to rounding of the decimal components.)

Step 4. Compute the dot product of the unit vectors:

$$\begin{aligned}\hat{a} \cdot \hat{b} &= 0.535(0.333) + (-0.267)(-0.667) + 0.802(0.667) \\ &= 0.178 + 0.178 + 0.535 = 0.891\end{aligned}$$

This matches the cosine similarity computed in Example 1, confirming the identity.

9. The Relationship Between Cosine Similarity and Euclidean Distance

For L2-normalized vectors, cosine similarity and Euclidean distance are related by a simple formula.

Claim. For unit vectors $\hat{a}$ and $\hat{b}$:

$$\|\hat{a} - \hat{b}\|^2 = 2(1 - \cos\theta)$$

Proof.

Step 1. Expand the squared Euclidean distance:

$$\begin{aligned}\|\hat{a} - \hat{b}\|^2 &= (\hat{a} - \hat{b}) \cdot (\hat{a} - \hat{b}) \\ &= \hat{a} \cdot \hat{a} - 2(\hat{a} \cdot \hat{b}) + \hat{b} \cdot \hat{b}\end{aligned}$$

Step 2. Since both vectors are unit vectors: $\hat{a} \cdot \hat{a} = 1$ and $\hat{b} \cdot \hat{b} = 1$. Also, $\hat{a} \cdot \hat{b} = \cos\theta$.

$$\|\hat{a} - \hat{b}\|^2 = 1 - 2\cos\theta + 1 = 2(1 - \cos\theta) \ \blacksquare$$

Consequences:

- When $\cos\theta = 1$ (identical direction): distance $= 0$. The vectors are the same point on the unit sphere.
- When $\cos\theta = 0$ (orthogonal): distance $= \sqrt{2} \approx 1.414$.
- When $\cos\theta = -1$ (opposite direction): distance $= 2$. The vectors are antipodal on the unit sphere.

Implication. On the unit hypersphere, maximizing cosine similarity is equivalent to minimizing Euclidean distance. Contrastive learning (Lecture 3) pushes matching pairs closer together on the unit sphere (higher cosine similarity, lower distance) and non-matching pairs further apart (lower cosine similarity, higher distance). Both formulations are mathematically equivalent after L2 normalization.

10. The Saturation Problem

There is an important limitation of cosine similarity that will matter for affinity calibration (Lecture 10).

The ceiling effect. Cosine similarity is bounded above by $+1$. This means that for two pairs of molecules — one binding with $K_d = 1$ nM (very tight) and another with $K_d = 0.01$ nM (100× tighter) — the cosine similarity compresses the distinction. As embeddings for strong binders converge toward the same direction, the score approaches 1.0 and cannot distinguish degrees of tightness.

Formally: The function $\cos\theta$ has derivative $-\sin\theta$, which approaches zero as $\theta \to 0$. Near $\theta = 0$ (high similarity), small changes in angle produce vanishingly small changes in cosine similarity:

$$\cos\theta \approx 1 - \frac{\theta^2}{2} \text{ for small } \theta$$

Two embeddings at $\theta = 5°$ have $\text{sim} = 0.9962$. Two embeddings at $\theta = 2°$ have $\text{sim} = 0.9994$. The 2.5× difference in angle (which may correspond to a 100× difference in binding affinity) produces only a 0.0032 difference in cosine similarity.

Consequence for affinity calibration. When we derive the affinity calibration framework in Lecture 10 ($\Delta G = ws + b$, where s is the contrastive score), this saturation creates a systematic nonlinearity at the high-affinity tail. The linear model may underpredict binding affinity for the tightest binders. This is a testable prediction that distinguishes given-bilinearity domains (where saturation may be less severe because the energy function is exact) from emergent-bilinearity domains (where the learned embedding may be more prone to compression).

11. Worked Example 4: Similarity Matrix with Normalized Vectors

Problem. Three antibodies have (unnormalized) embedding vectors in 2D:

$$q_1 = (3, 4), q_2 = (1, 0), q_3 = (-2, 1)$$

Two antigens have (unnormalized) embedding vectors in 2D:

$$k_1 = (2, 2), k_2 = (0, 5)$$

Compute the cosine similarity matrix.

Step 1. L2-normalize all vectors:

$$\| q_1 \| = \sqrt{9+16} = 5, \hat{q}_1 = (0.6, 0.8)$$

$$\| q_2 \| = \sqrt{1+0} = 1, \hat{q}_2 = (1, 0)$$

$$\| q_3 \| = \sqrt{4+1} = \sqrt{5} \approx 2.236, \hat{q}_3 = (-0.894, 0.447)$$

$$\| k_1 \| = \sqrt{4+4} = 2\sqrt{2} \approx 2.828, \hat{k}_1 = (0.707, 0.707)$$

$$\| k_2 \| = \sqrt{0+25} = 5, \hat{k}_2 = (0, 1)$$

Step 2. Compute $\widehat{Q}\widehat{K}^T$:

$$S_{11} = \hat{q}_1 \cdot \hat{k}_1 = 0.6(0.707) + 0.8(0.707) = 0.424 + 0.566 = 0.990$$

$$S_{12} = \hat{q}_1 \cdot \hat{k}_2 = 0.6(0) + 0.8(1) = 0.800$$

$$S_{21} = \hat{q}_2 \cdot \hat{k}_1 = 1(0.707) + 0(0.707) = 0.707$$

$$S_{22}=\hat{q}_2 \cdot \hat{k}_2=1(0)+0(1)=0$$
$$S_{31}=\hat{q}_3 \cdot \hat{k}_1=-0.894(0.707)+0.447(0.707)=-0.632+0.316=-0.316$$
$$S_{32}=\hat{q}_3 \cdot \hat{k}_2=-0.894(0)+0.447(1)=0.447$$

Step 3. Assemble the cosine similarity matrix:

$$S_{\cos}=\begin{pmatrix} 0.990 & 0.800 \\ 0.707 & 0 \\ -0.316 & 0.447 \end{pmatrix}$$

Step 4. Interpret:

	Ag 1	Ag 2
Ab 1	**0.990**	0.800
Ab 2	**0.707**	0
Ab 3	−0.316	**0.447**

Antibody 1 is highly compatible with both antigens, but slightly more with Antigen 1 (0.990 vs. 0.800). Antibody 2 is moderately compatible with Antigen 1 (0.707) but orthogonal to Antigen 2 (0) — completely unrelated. Antibody 3 is weakly compatible with Antigen 2 (0.447) and incompatible with Antigen 1 (−0.316).

Notice that every entry is in $[-1,+1]$ — the normalization ensures all scores are on the same scale regardless of the original magnitudes of the embedding vectors.

Comparison to raw dot products. Without normalization, the raw similarity matrix would be:

$$S_{\text{raw}}=Q K^T=\begin{pmatrix} 14 & 20 \\ 2 & 0 \\ -2 & 5 \end{pmatrix}$$

The raw dot product of $q_1 \cdot k_2=20$ is larger than $q_1 \cdot k_1=14$, suggesting Antibody 1 prefers Antigen 2. But this is an artifact of magnitude: Antigen 2 has a larger norm ($\| k_2 \|=5$ vs. $\| k_1 \|=2\sqrt{2}\approx 2.83$). After normalization, the correct ranking is reversed: Antibody 1 actually has higher angular alignment with Antigen 1 (0.990 vs. 0.800). L2 normalization prevents magnitude from distorting the ranking.

12. Summary

Concept	Definition	Role in MRC
Cosine similarity	$\text{sim}(a,b)=\dfrac{a \cdot b}{\| a \| \| b \|}=\cos\theta$	Magnitude-independent similarity measure
Range	$[-1,+1]$	Universal scale for all molecular comparisons
L2 normalization	$\hat{a}=a/\| a \|$	Projects vectors onto the unit hypersphere
Key identity	$\hat{a} \cdot \hat{b}=\cos\theta$	After normalization, dot product = cosine similarity
Distance relation	$\| \hat{a}-\hat{b} \|^2=2(1-\cos\theta)$	Maximizing similarity = minimizing distance
Saturation	$\cos\theta \to 1$ as $\theta \to 0$	Compresses high-affinity tail (Lecture 10)
Scale invariance	$\text{sim}(\alpha a,\beta b)=\text{sim}(a,b)$	Only direction matters, not magnitude

13. Checkpoint Questions

Q1. Compute the cosine similarity between $a=(1,1)$ and $b=(1,0)$.

Answer: $a \cdot b=1$. $\| a \|=\sqrt{2}$, $\| b \|=1$. $\text{sim}=1/\sqrt{2}\approx 0.707$, which corresponds to $\theta=45°$. The vector $(1,1)$ makes a 45° angle with the x-axis, consistent with this result.

Q2. Two molecular embeddings have cosine similarity 0.999. Another pair has cosine similarity 0.990. The first pair binds with $K_d=0.1$ nM and the second with $K_d=10$ nM (a 100-fold difference in affinity). Is cosine similarity a good predictor of binding affinity in this regime?

Answer: A 100-fold difference in affinity corresponds to only a 0.009 difference in cosine similarity. This is the saturation problem: cosine similarity is insensitive to affinity differences among tight binders. For retrieval (ranking which target an agent binds), this is acceptable — the ranking is preserved. For affinity prediction (estimating K_d or ΔG), the compression at the high-affinity end is problematic. This is why the affinity calibration framework (Lecture 10) tests for nonlinearity by fitting a quadratic model — the saturation would appear as a significant quadratic term.

Q3. If you L2-normalize all antibody and antigen embeddings, what constraint does this place on the similarity matrix $S = \hat{Q}\hat{K}^T$?

Answer: Every entry satisfies $-1 \leq S_{ij} \leq +1$. This is because each entry is a dot product of two unit vectors, which equals $\cos\theta_{ij} \in [-1, +1]$. The normalization guarantees that all scores are on a bounded, comparable scale — there are no entries with arbitrarily large magnitude. This is important for the softmax function (Derivation Guides 4 and 5): bounded inputs prevent softmax from saturating into a one-hot distribution.

Q4. An SFM produces embedding vectors q and k that are L2-normalized. The contrastive similarity score used in InfoNCE is $s = \hat{q} \cdot \hat{k} / \tau$. Why does dividing by temperature τ after normalization not reintroduce magnitude dependence?

Answer: Temperature τ is a global scalar that scales all scores equally — it does not depend on any individual vector's magnitude. After normalization, $\hat{q} \cdot \hat{k} = \cos\theta$ captures pure alignment. Dividing by τ controls the sharpness of the subsequent softmax (Derivation Guide 5) without reintroducing any vector-specific magnitude. Temperature is a property of the selection process (how decisive the choice is), not a property of any individual vector.

14. Connection to Future Lectures

- **Lecture 3:** CLIP and CALM both L2-normalize embeddings before computing the InfoNCE loss. The similarity matrix in contrastive learning is a cosine similarity matrix: $S_{ij} = \hat{z}_{x_i} \cdot \hat{z}_{y_j}$. Every entry is in $[-1, +1]$.
- **Lecture 7:** The SFM architecture prescribes L2 normalization as part of the projection head, following CLIP. This is not an arbitrary choice — it ensures that the contrastive score measures binding compatibility independent of encoder-specific magnitude artifacts.
- **Lecture 10:** The affinity calibration framework ($\Delta G = ws + b$) uses the contrastive score $s = \hat{q} \cdot \hat{k} / \tau$ as input. The saturation of cosine similarity at $+1$ predicts a systematic nonlinearity at the high-affinity tail. The two-parameter sufficiency test (linear vs. quadratic BIC comparison) diagnoses this nonlinearity. Given-bilinearity domains are predicted to show tighter linearity; emergent-bilinearity domains are predicted to show greater saturation effects.
- **Lecture 11:** The computational advantage of SFMs rests on the fact that $\hat{Q}\hat{K}^T$ computes all pairwise cosine similarities in one matrix multiplication. The L2 normalization step is $O(md)$ for m vectors of dimension d — negligible compared to the $O(mnd)$ cost of the matrix multiplication itself.